

# Extremal no-three-in-line subsets of a modular hyperbola in the Hall–Jackson–Sudbery–Wild window

Alex Komang\*

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## Abstract

Let  $p$  be an odd prime,  $c \in \mathbb{F}_p^*$ , and let  $H(c, p) = \{(x, y) \in \mathbb{Z}^2 : xy \equiv c \pmod{p}\}$  be the modular hyperbola. Hall, Jackson, Sudbery and Wild (1975) observed that the  $2p \times 2p$  window  $G(p) = [-\frac{p-1}{2}, \frac{3p-1}{2}] \times [0, 2p-1]$  contains exactly four points of each of the  $p-1$  residue classes of  $H(c, p)$ , at the vertices of a  $p \times p$  square, and that keeping three of the four (the twelve outer blocks  $T_2$  of the  $4 \times 4$  block subdivision of  $G(p)$ ) yields  $3(p-1)$  points with no three collinear. Kovács, Nagy and Szabó (2025/26) state that this deletion is optimal, i.e. that  $H(c, p) \cap T_2$  is a largest no-three-in-line subset of  $H(c, p) \cap G(p)$ . We give a short self-contained proof of this statement for every  $c$ , and we determine the complete extremal structure: we classify all lines carrying three or more points of  $H(c, p) \cap G(p)$  (they have slope  $\pm 1$ , and their number is  $\frac{3}{2}(p-1) - s$ ), we exhibit an explicit linear-programming certificate for the bound  $3(p-1)$  (so even the fractional relaxation has value  $3(p-1)$ ), and we show that the number of maximum no-three-in-line subsets is exactly  $9^s$ , where  $s \in \{0, 1, 2\}$  is the number of quadratic residues among  $c$  and  $-c$ . In particular the HJSW set is the unique maximum if and only if  $p \equiv 1 \pmod{4}$  and  $c$  is a quadratic non-residue. The bound does not depend on the position of the window: for *every*  $2p \times 2p$  box  $[x_0, x_0 + 2p) \times [y_0, y_0 + 2p)$  the modular hyperbola contains at most  $3(p-1)$  points in general position (an “orbit lemma”: in each orbit of the Klein four-group generated by  $(a, b) \mapsto (-b, -a)$  and  $(a, b) \mapsto (b, a)$  at most two of the four partner pairs are separated by the box), and the HJSW box is among the boxes where the bound is attained. The classification also gives the no-four-in-line analogue: the largest no-four-in-line subset of  $H(c, p) \cap G(p)$  has exactly  $\frac{7}{2}(p-1) + s$  points, so the set  $S_3$  of Kovács–Nagy–Szabó is optimal as well. Everything is verified by independent programs for all primes  $p \leq 101$  (all boxes for  $p \leq 61$ ).

## 1 Introduction

The no-three-in-line problem asks for the largest subset of the  $n \times n$  grid with no three collinear points; at most  $2n$  points are possible, and the best general lower bound,  $\frac{3}{2}n - o(n)$ , is due to Hall, Jackson, Sudbery and Wild [1]: for  $n = 2p$ ,  $p$  prime, they place  $3(p-1)$  points of a modular hyperbola  $H(c, p)$  in the window  $G(p)$ . Their construction rests on two facts: every line meets  $H(c, p)$  in at most two residue classes, and  $G(p)$  contains exactly four points of each class, at the vertices of a  $p \times p$  square; keeping the three copies outside the middle block of the  $4 \times 4$  block subdivision of  $G(p)$  leaves no three collinear.

Kovács, Nagy and Szabó [2] recently used the same window and its refinements for the no- $(k+1)$ -in-line problem. In their setting the HJSW set is the case  $|W| = 1$ ,  $k = 2$  of a general randomised construction (all four copies of every class are taken, then a minimum set  $R$  of points is deleted so that every line of slope  $\pm 1$  retains at most  $k$  points), and they state that for  $|W| = 1$ ,  $k = 2$  the HJSW deletion “essentially details an optimal choice” of the deleted points (v1, 2025), resp. yields the “largest no-3-in-line set  $S_2(c, p)$  inside  $S(W)$ ” (v2, 2026). The general

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\*Independent researcher, Bali, Indonesia. See the AI/tool-use disclosure at the end.

estimate of [2, Rem. 3.4],  $|R| \geq \frac{1}{2}X$  with  $X = \sum_{\ell}(|\ell \cap S(W)| - k)^+$ , gives for  $|W| = 1$ ,  $k = 2$  the bound  $|R| \geq p - 1 - s$  (see Remark 9 below), i.e.  $|S| \leq 3(p - 1) + s$  for every no-three-in-line  $S \subseteq H(c, p) \cap G(p)$ : exact when neither  $c$  nor  $-c$  is a square modulo  $p$ , but one or two short of the truth otherwise. We are not aware of a proof of the exact statement in [2]; the purpose of this note is to give one, and to determine the whole extremal structure of the problem.

**Results.** Throughout,  $p$  is an odd prime,  $h = (p - 1)/2$ ,  $c \in \mathbb{F}_p^*$ , and

$$H(c, p) = \{(x, y) \in \mathbb{Z}^2 : xy \equiv c \pmod{p}\}, \quad G(p) = \{-h, \dots, 3h + 1\} \times \{0, \dots, 2p - 1\},$$

and  $\mathcal{P} := H(c, p) \cap G(p)$ . A set is *lawful* (no-three-in-line, in general position) if it contains no three collinear points. A line is *rich* if it contains at least three points of  $\mathcal{P}$ ;  $\mathcal{L}$  denotes the set of rich lines. Let  $s = [c \text{ is a QR}] + [-c \text{ is a QR}] \in \{0, 1, 2\}$  (QR = quadratic residue modulo  $p$ ).

**Theorem 1.** (i) Every rich line has slope  $+1$  or  $-1$ ; there are exactly  $\frac{3}{2}(p - 1) - s$  rich lines,  $p - 1$  of them carrying three points of  $\mathcal{P}$  and  $\frac{1}{2}(p - 1) - s$  carrying four (Proposition 8 lists them); exactly  $2s$  points of  $\mathcal{P}$  lie on no rich line.

(ii) Every lawful subset of  $\mathcal{P}$  has at most  $3(p - 1)$  points; the bound is attained. Even the fractional relaxation (maximise  $\sum_q x_q$  subject to  $\sum_{q \in \ell} x_q \leq 2$  for  $\ell \in \mathcal{L}$ ,  $0 \leq x_q \leq 1$ ) has optimum  $3(p - 1)$ .

(iii) The number of lawful subsets of size  $3(p - 1)$  is exactly  $9^s$ . Every maximum set coincides with the HJSW set  $H(c, p) \cap T_2$  outside the (at most four) classes  $(a, \pm a)$  with  $a^2 = \pm c$ .

**Corollary 2.** If  $p \equiv 3 \pmod{4}$  then  $s = 1$  for every  $c$ : there are exactly 9 maximum sets. If  $p \equiv 1 \pmod{4}$  then  $s = 2$  for  $c$  a QR (81 maximum sets) and  $s = 0$  for  $c$  a non-residue: the HJSW set is the unique maximum. Thus  $H(c, p) \cap T_2$  is the unique largest lawful subset of  $H(c, p) \cap G(p)$  if and only if  $p \equiv 1 \pmod{4}$  and  $c$  is a quadratic non-residue.

**Theorem 3.** Let  $x_0, y_0 \in \mathbb{Z}$  and  $W = [x_0, x_0 + 2p) \times [y_0, y_0 + 2p)$ . Every lawful subset of  $H(c, p) \cap W$  has at most  $3(p - 1)$  points.

The proof of Theorem 1 is a double count over the rich lines: for a lawful  $S$ ,  $|S| \leq Z + 2|\mathcal{L}|$  where  $Z$  is the number of points on no rich line, and the identity  $Z + 2|\mathcal{L}| = 3(p - 1)$  follows from the classification of the rich lines. The count is exact precisely because the  $2s$  points that lie on no rich line are accounted separately; this is the correction to the averaging bound of [2] mentioned above. All rich lines lie inside orbits of a Klein four-group acting on the classes, and the equality analysis reduces to two small “gadgets” (Figure 1), which gives  $9^s$ . For a general box the same count gives  $Z + 2|\mathcal{L}| = 2(p - 1) + 2s + 2\sum_O(e_O + f_O)$ , where  $e_O, f_O \leq 2$  count the partner pairs of the orbit  $O$  that are “split” by the box; the orbit lemma  $e_O + f_O \leq 2$  (Section 5) gives Theorem 3. For boxes other than the HJSW box the maximum can be smaller than  $3(p - 1)$  (Table 1). The four-point rich lines are pairwise disjoint, which settles the no-four-in-line analogue as well (Corollary 11: the maximum is  $\frac{7}{2}(p - 1) + s$  in the HJSW box, attained by the set  $S_3$  of [2]).

## 2 Notation

Two points are *congruent* if their coordinates agree modulo  $p$ ; a *class* is a congruence class, identified with an element of  $\mathbb{F}_p \times \mathbb{F}_p$ . Points of  $H(c, p)$  have both coordinates prime to  $p$ , so the classes of  $H(c, p)$  are the  $p - 1$  pairs  $\kappa = (a, b) \in \mathbb{F}_p^* \times \mathbb{F}_p^*$  with  $ab = c$ . Since each coordinate range of  $G(p)$  is an interval of  $2p$  consecutive integers, it contains exactly two representatives of every residue class; hence every class  $\kappa$  of  $H(c, p)$  has exactly four points in  $G(p)$  and  $|\mathcal{P}| = 4(p - 1)$ . We

write  $(x_\kappa, y_\kappa)$  for the *base copy* of  $\kappa$ , its unique point in  $\{-h, \dots, h\} \times \{1, \dots, p-1\}$  (necessarily  $x_\kappa \neq 0$ ), and

$$\kappa(r, s) := (x_\kappa + rp, y_\kappa + sp), \quad r, s \in \{0, 1\},$$

for its four points in  $G(p)$ . Following [1, 2] we sort the classes into four *types* according to the position of the base copy:

$$A : x_\kappa > 0, y_\kappa > h; \quad B : x_\kappa < 0, y_\kappa > h; \quad C : x_\kappa > 0, y_\kappa \leq h; \quad D : x_\kappa < 0, y_\kappa \leq h.$$

(In the notation of [2, Def. 2.3]:  $\kappa(0, 0) \in A_{0,0}, B_{-1,0}, C_{0,0}, D_{-1,0}$  respectively; the HJSW set  $T_2$  omits, for each type, the copy  $A_{0,0}, B_{0,0}, C_{0,1}, D_{0,1}$ , i.e.  $\kappa(0, 0)$  for  $A$ ,  $\kappa(1, 0)$  for  $B$ ,  $\kappa(0, 1)$  for  $C$ ,  $\kappa(1, 1)$  for  $D$ ; the union of these omitted blocks is the middle block  $M$  of [2, §2.3]. We call this copy the *middle copy* of  $\kappa$ .) We write  $n_A, n_B, n_C, n_D$  for the numbers of classes of each type, so  $n_A + n_B + n_C + n_D = p - 1$ . Finally, for a class  $\kappa$  put

$$d_\kappa := x_\kappa - y_\kappa, \quad e_\kappa := x_\kappa + y_\kappa, \quad D_\kappa := \{x - y = d_\kappa\}, \quad E_\kappa := \{x + y = e_\kappa + p\};$$

$D_\kappa$  (the *diagonal* of  $\kappa$ ) contains  $\kappa(0, 0), \kappa(1, 1)$  and  $E_\kappa$  (the *antidiagonal*) contains  $\kappa(0, 1), \kappa(1, 0)$ . For a real number  $x$  let  $x^+ = \max\{x, 0\}$ .

Three involutions act on the classes of  $H(c, p)$ :

$$\sigma(a, b) = (-b, -a), \quad \tau(a, b) = (b, a), \quad \nu(a, b) = (-a, -b) = \sigma\tau(a, b) = \tau\sigma(a, b),$$

so  $V = \{1, \sigma, \tau, \nu\}$  is a Klein four-group acting on the  $p - 1$  classes.  $\sigma$  has fixed classes iff  $-c$  is a QR (then exactly two:  $(a, -a)$  with  $a^2 = -c$ ),  $\tau$  has fixed classes iff  $c$  is a QR (two:  $(a, a)$  with  $a^2 = c$ ), and  $\nu$  has none. Put  $f_\sigma := \#\{\kappa : \sigma\kappa = \kappa\} \in \{0, 2\}$ ,  $f_\tau := \#\{\kappa : \tau\kappa = \kappa\} \in \{0, 2\}$ ; thus  $f_\sigma + f_\tau = 2s$ . The  $V$ -orbits are of size 4 (*generic*: four distinct classes) or of size 2 (*exceptional*:  $\{\kappa, \nu\kappa\}$  with  $\kappa$  and  $\nu\kappa$  both fixed by  $\sigma$ , or both fixed by  $\tau$ ); there are exactly  $s$  exceptional orbits.

### 3 Rich lines

**Lemma 4.** *Let  $\ell$  be a line containing at least two points of  $H(c, p)$ . Then the points of  $H(c, p) \cap \ell$  lie in at most two classes. More precisely, if  $\ell$  contains a point of class  $\kappa$ , then the classes met by  $\ell$  are: only  $\kappa$  if  $\ell$  has slope 0 or  $\infty$ ; among  $\{\kappa, \sigma\kappa\}$  if  $\ell$  has slope  $+1$ ; among  $\{\kappa, \tau\kappa\}$  if  $\ell$  has slope  $-1$ .*

*Proof.* Let  $(u, v) \in \mathbb{Z}^2$  be a primitive direction vector of  $\ell$  and  $(x_0, y_0) \in H(c, p) \cap \ell$ . The lattice points of  $\ell$  are  $(x_0 + tu, y_0 + tv)$ ,  $t \in \mathbb{Z}$ , and such a point lies on  $H(c, p)$  iff  $uv t^2 + (x_0 v + y_0 u) t \equiv 0 \pmod{p}$  (we used  $x_0 y_0 \equiv c$ ). The class of the point depends only on  $t \pmod{p}$ . If  $uv \not\equiv 0$ , the congruence has at most two roots modulo  $p$ ; if exactly one of  $u, v$  is divisible by  $p$  (both cannot be, as  $\gcd(u, v) = 1$ ), it reduces to  $x_0 v t \equiv 0$  or  $y_0 u t \equiv 0$  with a unit coefficient, so  $t \equiv 0$ : one class. For  $(u, v) = (1, 0), (0, 1)$  this gives the claim for slopes 0,  $\infty$ . For  $(u, v) = (1, 1)$  the roots are  $t \equiv 0$  and  $t \equiv -(x_0 + y_0)$ , the latter giving the point  $(-y_0, -x_0)$  modulo  $p$ , of class  $\sigma\kappa$ . For  $(u, v) = (1, -1)$  the roots are  $t \equiv 0$  and  $t \equiv y_0 - x_0$ , giving  $(y_0, x_0)$ , of class  $\tau\kappa$ .  $\square$

**Lemma 5.** *Two distinct points of one class in  $G(p)$  differ by  $\pm(p, 0)$ ,  $\pm(0, p)$ ,  $\pm(p, p)$  or  $\pm(p, -p)$ ; the line through them has slope 0,  $\infty$ ,  $+1$  or  $-1$  and contains no third point of that class. The point  $\kappa(r, s)$  lies on the slope- $(+1)$  line  $x - y = d_\kappa + (r - s)p$  and on the slope- $(-1)$  line  $x + y = e_\kappa + (r + s)p$ . In particular  $D_\kappa$  and  $E_\kappa$  are the only lines of slope  $\pm 1$  carrying two points of  $\kappa$ .*

*Proof.* The four points of a class are the vertices of an axis-parallel square of side  $p$ ; no three are collinear. The rest is a direct computation.  $\square$

**Corollary 6.** *A rich line has slope +1 or -1. Consequently a set  $S \subseteq \mathcal{P}$  is lawful iff  $|S \cap \ell| \leq 2$  for every  $\ell \in \mathcal{L}$ .*

*Proof.* By Lemma 4 three points of  $\mathcal{P}$  on a line lie in at most two classes, so two of them are congruent, and by Lemma 5 the line has slope 0,  $\infty$ ,  $\pm 1$ . A line of slope 0 or  $\infty$  meets a single class (Lemma 4), which has at most two points on it (Lemma 5).  $\square$

**Lemma 7** (position of the partner classes). *For every class  $\kappa$  the types of  $\sigma\kappa$ ,  $\tau\kappa$  and the shifts  $d_{\sigma\kappa} - d_\kappa$ ,  $e_{\tau\kappa} - e_\kappa$  are given by the following table:*

| type of $\kappa$ | type of $\sigma\kappa$ | $d_{\sigma\kappa} - d_\kappa$ | type of $\tau\kappa$ | $e_{\tau\kappa} - e_\kappa$ |
|------------------|------------------------|-------------------------------|----------------------|-----------------------------|
| $A$              | $A$                    | 0                             | $D$                  | $-p$                        |
| $B$              | $C$                    | $+p$                          | $B$                  | 0                           |
| $C$              | $B$                    | $-p$                          | $C$                  | 0                           |
| $D$              | $D$                    | 0                             | $A$                  | $+p$                        |

*In particular  $\sigma$  preserves the types  $A$ ,  $D$  and interchanges  $B$ ,  $C$ ;  $\tau$  preserves  $B$ ,  $C$  and interchanges  $A$ ,  $D$ ;  $\nu$  interchanges  $A \leftrightarrow D$  and  $B \leftrightarrow C$ ; the fixed classes of  $\sigma$  lie in  $A \cup D$ , those of  $\tau$  in  $B \cup C$ ; and  $n_A = n_D$ ,  $n_B = n_C$ .*

*Proof.* Write  $\kappa = (a, b)$  with base copy  $(x_\kappa, y_\kappa)$ , so  $x_\kappa \equiv a$ ,  $y_\kappa \equiv b$ . The base copy of  $\sigma\kappa = (-b, -a)$  has first coordinate  $-y_\kappa$  if  $y_\kappa \leq h$  and  $p - y_\kappa$  if  $y_\kappa > h$ , and second coordinate  $p - x_\kappa$  if  $x_\kappa > 0$  and  $-x_\kappa$  if  $x_\kappa < 0$  (these are the representatives of  $-b$  in  $[-h, h]$  and of  $-a$  in  $[1, p-1]$ ). Going through the four types:  $\kappa \in A$ :  $(p - y_\kappa, p - x_\kappa)$ , again of type  $A$ , difference  $x_\kappa - y_\kappa$ ;  $\kappa \in D$ :  $(-y_\kappa, -x_\kappa)$ , type  $D$ , difference  $x_\kappa - y_\kappa$ ;  $\kappa \in C$ :  $(-y_\kappa, p - x_\kappa)$ , type  $B$ , difference  $x_\kappa - y_\kappa - p$ ;  $\kappa \in B$ :  $(p - y_\kappa, -x_\kappa)$ , type  $C$ , difference  $x_\kappa - y_\kappa + p$ . Similarly, the base copy of  $\tau\kappa = (b, a)$  has first coordinate  $y_\kappa$  if  $y_\kappa \leq h$  and  $y_\kappa - p$  otherwise, and second coordinate  $x_\kappa$  if  $x_\kappa > 0$  and  $x_\kappa + p$  otherwise:  $\kappa \in B$ :  $(y_\kappa - p, x_\kappa + p)$ , type  $B$ , sum  $e_\kappa$ ;  $\kappa \in C$ :  $(y_\kappa, x_\kappa)$ , type  $C$ , sum  $e_\kappa$ ;  $\kappa \in A$ :  $(y_\kappa - p, x_\kappa)$ , type  $D$ , sum  $e_\kappa - p$ ;  $\kappa \in D$ :  $(y_\kappa, x_\kappa + p)$ , type  $A$ , sum  $e_\kappa + p$ . The remaining assertions follow.  $\square$

**Proposition 8.**  $\mathcal{L}$  consists of the following lines, and no others; for each we list its points in  $\mathcal{P}$ .

- (a) For  $\kappa \in A \cup D$  with  $\sigma\kappa \neq \kappa$ : the common diagonal  $D_\kappa = D_{\sigma\kappa}$ , with the four points  $\kappa(0, 0), \kappa(1, 1), \sigma\kappa(0, 0), \sigma\kappa(1, 1)$ .
- (b) For  $\kappa \in B \cup C$  with  $\tau\kappa \neq \kappa$ : the common antidiagonal  $E_\kappa = E_{\tau\kappa}$ , with the four points  $\kappa(0, 1), \kappa(1, 0), \tau\kappa(0, 1), \tau\kappa(1, 0)$ .
- (c) For every  $\kappa \in C$ :  $D_\kappa$  with the three points  $\kappa(0, 0), \kappa(1, 1), \sigma\kappa(1, 0)$  ( $\sigma\kappa \in B$ ); for every  $\kappa \in B$ :  $D_\kappa$  with the three points  $\kappa(0, 0), \kappa(1, 1), \sigma\kappa(0, 1)$  ( $\sigma\kappa \in C$ ).
- (d) For every  $\kappa \in A$ :  $E_\kappa$  with the three points  $\kappa(0, 1), \kappa(1, 0), \tau\kappa(1, 1)$  ( $\tau\kappa \in D$ ); for every  $\kappa \in D$ :  $E_\kappa$  with the three points  $\kappa(0, 1), \kappa(1, 0), \tau\kappa(0, 0)$  ( $\tau\kappa \in A$ ).

The lines listed in (a)–(d) are pairwise distinct (apart from the identifications  $D_\kappa = D_{\sigma\kappa}$  in (a) and  $E_\kappa = E_{\tau\kappa}$  in (b) that are built into the list), and each of them lies inside a single  $V$ -orbit of classes. Consequently

$$|\mathcal{L}| = \frac{n_A + n_D - f_\sigma}{2} + \frac{n_B + n_C - f_\tau}{2} + (n_A + n_D) + (n_B + n_C) = \frac{3}{2}(p-1) - s, \quad (1)$$

with  $p-1$  three-point lines and  $\frac{1}{2}(p-1) - s$  four-point lines. Moreover, the points of  $\mathcal{P}$  lying on no rich line are exactly  $\kappa(1, 1)$  for  $\sigma$ -fixed  $\kappa \in A$ ,  $\kappa(0, 0)$  for  $\sigma$ -fixed  $\kappa \in D$ ,  $\kappa(0, 1)$  for  $\tau$ -fixed  $\kappa \in B$  and  $\kappa(1, 0)$  for  $\tau$ -fixed  $\kappa \in C$ ; their number is

$$Z := \#\{q \in \mathcal{P} : q \text{ lies on no rich line}\} = f_\sigma + f_\tau = 2s. \quad (2)$$

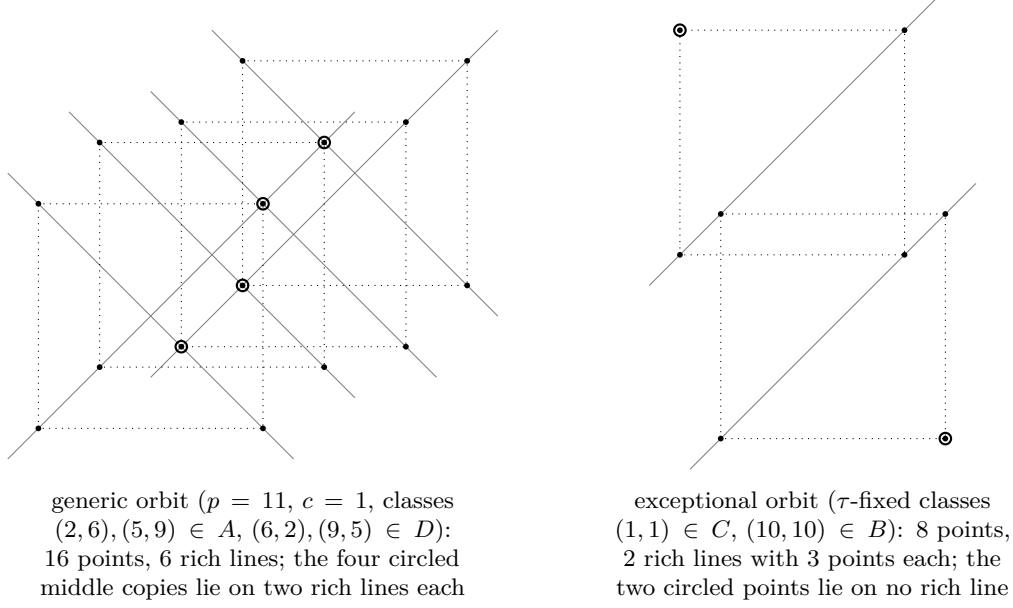


Figure 1: The two gadgets of the equality analysis, drawn to scale for  $p = 11, c = 1$  (dotted squares: the four copies of a class; grey: rich lines). Generic orbit: the circled points are excluded and the other 12 are forced ( $= T_2$ ). Exceptional orbit: the circled points are forced and each rich line keeps two of its three points,  $3 \cdot 3 = 9$  choices.

*The points lying on two rich lines are exactly the middle copies of the classes  $\kappa$  whose partner ( $\sigma\kappa$  for  $\kappa \in A \cup D, \tau\kappa$  for  $\kappa \in B \cup C$ ) is different from  $\kappa$ ; no point lies on three rich lines.*

*Proof.* Let  $\ell \in \mathcal{L}$ . By Corollary 6 its slope is  $\pm 1$ ; say  $+1$  (the case  $-1$  is symmetric, with  $\tau, e_\kappa, E_\kappa$  in place of  $\sigma, d_\kappa, D_\kappa$ ). By Lemma 4 the points of  $\ell \cap \mathcal{P}$  lie in two classes  $\kappa, \sigma\kappa$  (a single class carries at most two points on a line), and one of the classes, say  $\kappa$ , has two points on  $\ell$ ; by Lemma 5 these are  $\kappa(0, 0), \kappa(1, 1)$  and  $\ell = D_\kappa$ , and  $\sigma\kappa \neq \kappa$ . The points of  $\sigma\kappa$  on  $D_\kappa$  are the  $\sigma\kappa(r, s)$  with  $d_{\sigma\kappa} + (r - s)p = d_\kappa$ . By Lemma 7: for  $\kappa \in A \cup D, r = s$ , giving (a); for  $\kappa \in C, r - s = 1$ , giving  $\sigma\kappa(1, 0)$ ; for  $\kappa \in B, r - s = -1$ , giving  $\sigma\kappa(0, 1)$ ; this is (c). Conversely each listed line has at least three points.

Distinctness: if  $D_\kappa = D_\lambda$  then this line of slope  $+1$  meets both classes  $\kappa, \lambda$ , so  $\lambda \in \{\kappa, \sigma\kappa\}$  by Lemma 4; and  $D_\kappa = D_{\sigma\kappa}$  holds for  $\kappa \in A \cup D$  (shift 0) but not for  $\kappa \in B \cup C$  (shift  $\pm p$ ). Likewise  $E_\kappa = E_\lambda$  forces  $\lambda \in \{\kappa, \tau\kappa\}$ , and  $E_\kappa = E_{\tau\kappa}$  holds exactly for  $\kappa \in B \cup C$ . A rich line meets only the classes  $\kappa, \sigma\kappa$  (or  $\kappa, \tau\kappa$ ), which lie in one  $V$ -orbit. Counting: (a) gives  $(n_A + n_D - f_\sigma)/2$  lines, (b)  $(n_B + n_C - f_\tau)/2$ , (c)  $n_B + n_C$ , (d)  $n_A + n_D$ , which is (1).

For the incidences of a single point, take  $\kappa \in A$ . Its points  $\kappa(0, 1), \kappa(1, 0)$  lie on  $E_\kappa \in \mathcal{L}$  (d), and on no line of type (a)/(c) (their slope- $(+1)$  lines  $x - y = d_\kappa \pm p$  carry no second point of  $\kappa$  and, by the table, no point of  $\sigma\kappa$ ).  $\kappa(1, 1)$  lies on  $D_\kappa$ , which is rich iff  $\sigma\kappa \neq \kappa$ , and on  $x + y = e_\kappa + 2p = e_{\tau\kappa} + 3p$ , which carries no point of  $\tau\kappa$ : so  $\kappa(1, 1)$  is on one rich line if  $\sigma\kappa \neq \kappa$  and on none otherwise.  $\kappa(0, 0)$  lies on  $D_\kappa$  and on  $x + y = e_\kappa = e_{\tau\kappa} + p$ , i.e. on  $E_{\tau\kappa} \in \mathcal{L}$ : two rich lines if  $\sigma\kappa \neq \kappa$ , one otherwise. The types  $D, B, C$  are analogous (for  $B$ :  $\kappa(1, 0) \in D_{\sigma\kappa}$  since  $d_\kappa + p = d_{\sigma\kappa}$ , while  $\kappa(0, 1)$  lies on  $x - y = d_\kappa - p = d_{\sigma\kappa} - 2p$ , which carries no point of  $\sigma\kappa$ ; for  $C$  the roles of  $(1, 0), (0, 1)$  are exchanged; for  $D$ :  $\kappa(1, 1) \in E_{\tau\kappa}$  since  $e_\kappa + 2p = e_{\tau\kappa} + p$ , and  $\kappa(0, 0)$  lies on  $x + y = e_\kappa = e_{\tau\kappa} - p$ ). This gives (2) and the last two assertions.  $\square$

## 4 Proof of Theorem 1 and Corollary 2

Part (i) is Corollary 6 and Proposition 8.

*Upper bound.* Let  $S \subseteq \mathcal{P}$  be lawful. For  $q \in \mathcal{P}$  let  $m(q)$  be the number of rich lines through  $q$ . Every rich line carries at most two points of  $S$  (Corollary 6), hence

$$\begin{aligned} |S| &= \#\{q \in S : m(q) = 0\} + \#\{q \in S : m(q) \geq 1\} \leq Z + \sum_{q \in S} m(q) - \sum_{q \in S} (m(q) - 1)^+ \\ &= Z + \sum_{\ell \in \mathcal{L}} |S \cap \ell| - \sum_{q \in S} (m(q) - 1)^+ \leq Z + 2|\mathcal{L}|, \end{aligned} \quad (3)$$

and by (1), (2),  $Z + 2|\mathcal{L}| = 2s + 3(p - 1) - 2s = 3(p - 1)$ . The HJSW set attains the bound [1].

*LP certificate.* The same count is valid for fractional  $x \in [0, 1]^{\mathcal{P}}$  with  $\sum_{q \in \ell} x_q \leq 2$  ( $\ell \in \mathcal{L}$ ):  $\sum_q x_q \leq \sum_{q: m(q)=0} x_q + \sum_{\ell \in \mathcal{L}} \sum_{q \in \ell} x_q \leq Z + 2|\mathcal{L}|$ . In LP-duality terms the dual solution assigns weight 1 to every rich line and weight 1 to the bound  $x_q \leq 1$  of each of the  $Z$  points on no rich line. This proves (ii).

*Equality.*  $|S| = 3(p - 1)$  holds iff all inequalities in (3) are equalities, i.e. iff (I)  $S$  contains all  $Z$  points lying on no rich line; (II)  $|S \cap \ell| = 2$  for every  $\ell \in \mathcal{L}$ ; (III) no point of  $S$  lies on two rich lines. Since every rich line lies inside one  $V$ -orbit of classes (Proposition 8), conditions (I)–(III) split into independent conditions on the orbits, and the number of maximum sets is the product of the numbers of admissible choices in the orbits. In a generic orbit  $\{\kappa, \sigma\kappa, \tau\kappa, \nu\kappa\}$  all four classes have distinct partners; the six rich lines meeting the orbit (two of type (a) or (b), four of type (c) or (d)) meet no other orbit, its four middle copies have  $m = 2$  and are excluded by (III), and each of the six lines then retains exactly two points, both with  $m = 1$ , which are forced into  $S$  by (II): on a generic orbit  $S$  coincides with  $H(c, p) \cap T_2$ , and there is exactly one admissible choice. An exceptional orbit  $\{\kappa, \nu\kappa\}$  ( $\kappa$  and  $\nu\kappa$  fixed by  $\sigma$ , or both fixed by  $\tau$ ) meets exactly two rich lines, each with three points of  $m = 1$ , and contains two points with  $m = 0$ : conditions (I)–(III) allow  $\binom{3}{2}\binom{3}{2} = 9$  choices, and all of them are lawful by Corollary 6. There are  $s$  exceptional orbits, whence  $9^s$  maximum sets, all agreeing with the HJSW set outside the exceptional orbits, i.e. outside the classes  $(a, \pm a)$  with  $a^2 = \pm c$ . This proves (iii).

*Corollary 2.*  $-1$  is a QR modulo  $p$  iff  $p \equiv 1 \pmod{4}$ . If  $p \equiv 3 \pmod{4}$ , exactly one of  $c, -c$  is a QR, so  $s = 1$ ; if  $p \equiv 1 \pmod{4}$ ,  $c$  and  $-c$  have the same character, so  $s \in \{0, 2\}$  according as  $c$  is a non-residue or a residue.  $\square$

*Remark 9.* (a) In the language of [2, Constr. 3.2, Rem. 3.4] with  $W = \{c\}$ ,  $k = 2$  (so  $S(W) = \mathcal{P}$  and  $L'$  is the set of lines of slope  $\pm 1$ ):  $X_{2,p}(\{c\}) = \sum_{\ell} (|\ell \cap \mathcal{P}| - 2)^+ = (p - 1) \cdot 1 + (\frac{1}{2}(p - 1) - s) \cdot 2 = 2(p - 1) - 2s$  by Theorem 1(i), so the general bound  $|R| \geq \frac{1}{2}X$  gives  $|R| \geq p - 1 - s$  and hence, for every lawful  $S \subseteq \mathcal{P}$ ,  $|S| \leq |\mathcal{P}| - |R| \leq 3(p - 1) + s$  (a lawful  $S$  is a trimming, by Corollary 6). The averaging loses exactly one point per exceptional orbit, because each such orbit contains two points on no rich line and no point on two rich lines; the count (3) keeps track of these points and is exact.

(b) The theorem says that no lawful set can use the middle block  $M$  profitably: adding fourth copies to the HJSW set is impossible without removing at least as many points. Any improvement of the HJSW construction must therefore use points outside  $H(c, p) \cap G(p)$  (another hyperbola, a larger window — by Theorem 3 not merely a shifted  $2p \times 2p$  box — or non-hyperbola points).

(c) The proof uses only: at most two classes per line (Lemma 4), the square arrangement of the four copies, and the position of the partner classes (Lemma 7). It does not use  $p \geq 7$ ;  $p = 3$  and  $p = 5$  are covered as well.

## 5 All $2p \times 2p$ boxes

Let  $W = [x_0, x_0 + 2p) \times [y_0, y_0 + 2p)$  with  $x_0, y_0 \in \mathbb{Z}$  (only  $x_0, y_0$  modulo  $p$  matter),  $\mathcal{P}_W = H(c, p) \cap W$ , and let  $\rho_x(u) \in [x_0, x_0 + p)$ ,  $\rho_y(u) \in [y_0, y_0 + p)$  denote the representatives of a residue  $u$ . Every class  $\kappa = (a, b)$  of  $H(c, p)$  again has exactly four points  $\kappa(r, s) = (\rho_x(a) + rp, \rho_y(b) + sp)$ ,  $r, s \in \{0, 1\}$ , in  $W$ ; put  $d_\kappa = \rho_x(a) - \rho_y(b)$ ,  $e_\kappa = \rho_x(a) + \rho_y(b)$ ,  $D_\kappa = \{x - y = d_\kappa\}$ ,  $E_\kappa =$

$\{x + y = e_\kappa + p\}$ . Lemmas 4, 5 and Corollary 6 hold verbatim (they do not use the position of the box), so a rich line is  $D_\kappa$  for a class with  $\sigma\kappa \neq \kappa$  or  $E_\kappa$  for a class with  $\tau\kappa \neq \kappa$ , and its points are the two copies  $\kappa(0, 0), \kappa(1, 1)$  (resp.  $\kappa(0, 1), \kappa(1, 0)$ ) together with the copies of the partner class on the same line. Since  $d_\kappa \equiv d_{\sigma\kappa} \pmod{p}$  and both lie in an interval of length  $2p - 2$ ,  $d_{\sigma\kappa} - d_\kappa \in \{-p, 0, p\}$ ; we call the  $\sigma$ -pair  $\{\kappa, \sigma\kappa\}$  *shared* if  $d_{\sigma\kappa} = d_\kappa$  (then  $D_\kappa = D_{\sigma\kappa}$  carries the four points  $\kappa(0, 0), \kappa(1, 1), \sigma\kappa(0, 0), \sigma\kappa(1, 1)$ ) and *split* otherwise (then  $D_\kappa$  carries  $\kappa(0, 0), \kappa(1, 1)$  and one copy of  $\sigma\kappa$ , and  $D_{\sigma\kappa}$  carries  $\sigma\kappa(0, 0), \sigma\kappa(1, 1)$  and one copy of  $\kappa$ ). Likewise a  $\tau$ -pair is shared ( $E_\kappa = E_{\tau\kappa}$ , four points) or split (two lines with three points). In the HJSW box the  $\sigma$ -pairs in  $A \cup D$  and the  $\tau$ -pairs in  $B \cup C$  are shared and the others are split (Lemma 7); in the box  $[0, 2p]^2$  all pairs are shared.

**Lemma 10** (orbit lemma). *Let  $O = \{\kappa, \sigma\kappa, \tau\kappa, \nu\kappa\}$  be a generic orbit, and let  $e_O$  (resp.  $f_O$ ) be the number of split  $\sigma$ -pairs (resp.  $\tau$ -pairs) among the two  $\sigma$ -pairs and two  $\tau$ -pairs of  $O$ . Then  $e_O + f_O \in \{0, 2\}$ ; in particular  $e_O + f_O \leq 2$ .*

*Proof.* Write  $\kappa = (a, b)$ ,  $X_u = \rho_x(u)$ ,  $X'_u = \rho_x(-u)$ ,  $Y_u = \rho_y(u)$ ,  $Y'_u = \rho_y(-u)$  for  $u \in \{a, b\}$ . The base copies are  $P = (X_a, Y_b)$  of  $\kappa$ ,  $Q = (X'_b, Y'_a)$  of  $\sigma\kappa$ ,  $R = (X_b, Y_a)$  of  $\tau\kappa$  and  $T = (X'_a, Y'_b)$  of  $\nu\kappa$ ; the  $\sigma$ -pairs are  $\{P, Q\}$ ,  $\{R, T\}$  and the  $\tau$ -pairs  $\{P, R\}$ ,  $\{Q, T\}$ . With  $d(x, y) = x - y$ ,  $e(x, y) = x + y$ , the four split conditions are  $E_1 : d_P \neq d_Q$ ,  $E_2 : d_T \neq d_R$ ,  $F_1 : e_P \neq e_R$ ,  $F_2 : e_Q \neq e_T$ ; each of the four differences is  $\equiv 0 \pmod{p}$  and lies in  $[-p, p]$ , hence is one of  $-p, 0, p$ .

Let  $m_x$  be the least multiple of  $p$  exceeding  $2x_0$  and  $\mu_x = m_x - 2x_0 \in [1, p]$ ; similarly  $m_y, \mu_y$ . Since  $X_u + X'_u \equiv 0 \pmod{p}$  and  $X_u + X'_u \in [2x_0 + 1, 2x_0 + 2p - 3]$ , we have  $X_u + X'_u = m_x + \varepsilon_x(u)p$  with  $\varepsilon_x(u) \in \{0, 1\}$ , and explicitly  $\varepsilon_x(u) = [X_u - x_0 > \mu_x]$ ; likewise  $Y_u + Y'_u = m_y + \varepsilon_y(u)p$ ,  $\varepsilon_y(u) = [Y_u - y_0 > \mu_y]$ . Put  $\Delta = \varepsilon_x(a) - \varepsilon_x(b)$ ,  $\Gamma = \varepsilon_y(a) - \varepsilon_y(b)$ ,  $A = \varepsilon_x(b) - \varepsilon_y(a)$ ,  $B = \varepsilon_x(a) - \varepsilon_y(b)$ , and define the integers  $\theta, \eta$  by

$$(X_a + X_b) - (Y_a + Y_b) - (m_x - m_y) = \theta p, \quad (X_b - X_a) - (Y_b - Y_a) = \eta p.$$

Substituting  $X'_u = m_x + \varepsilon_x(u)p - X_u$ ,  $Y'_u = m_y + \varepsilon_y(u)p - Y_u$  one finds

$$d_P - d_Q = (\theta - A)p, \quad d_T - d_R = (B - \theta)p, \quad e_R - e_P = \eta p, \quad e_Q - e_T = (\Gamma - \Delta - \eta)p,$$

so that  $E_1 \Leftrightarrow \theta \neq A$ ,  $E_2 \Leftrightarrow \theta \neq B$ ,  $F_1 \Leftrightarrow \eta \neq 0$ ,  $F_2 \Leftrightarrow \eta \neq \Gamma - \Delta$ , and  $\theta - A, \theta - B, \eta, \eta - (\Gamma - \Delta) \in \{-1, 0, 1\}$ .

*Parity.* For  $x \in \{-1, 0, 1\}$ ,  $[x \neq 0] \equiv x \pmod{2}$ . Hence  $E_1 + E_2 + F_1 + F_2 \equiv (\theta - A) + (\theta - B) + \eta + (\eta - \Gamma + \Delta) \equiv A + B + \Gamma - \Delta = 2\varepsilon_x(b) - 2\varepsilon_y(b) \equiv 0 \pmod{2}$ : the number of split pairs in  $O$  is even.

If  $F_1$  holds, then  $E_1$  and  $E_2$  do not both hold. Put  $s = X_a - x_0$ ,  $t = X_b - x_0 \in [0, p - 1]$  and  $v = (x_0 - y_0) \pmod{p}$ ; since  $Y_u \equiv X_u$ , we have  $Y_a - y_0 = s + v - c_a p$  and  $Y_b - y_0 = t + v - c_b p$  with  $c_a = [s + v \geq p]$ ,  $c_b = [t + v \geq p]$ , and therefore  $\eta = c_b - c_a$ . So  $F_1$  means  $c_a \neq c_b$ ; the substitution  $a \leftrightarrow b$  exchanges  $E_1 \leftrightarrow E_2$  and fixes  $F_1$ , so we may assume  $c_a = 0$ ,  $c_b = 1$ , i.e.  $s \leq p - 1 - v < t$ . Since  $\mu_x - \mu_y \equiv -2v \pmod{p}$  and  $|\mu_x - \mu_y| \leq p - 1$ , we have  $\mu_x - \mu_y = -2v + kp$  with  $k \in \{0, 1, 2\}$ , and a direct computation gives  $\theta = 1 - k$ . With  $\varepsilon_x(a) = [s > \mu_x]$ ,  $\varepsilon_x(b) = [t > \mu_x]$ ,  $\varepsilon_y(a) = [s + v > \mu_y]$ ,  $\varepsilon_y(b) = [t + v - p > \mu_y]$  and  $\mu_y = \mu_x + 2v - kp$ :

$k = 1$ :  $E_1 \Leftrightarrow [t > \mu_x] \neq [s > \mu_x + v - p]$  and  $E_2 \Leftrightarrow [s > \mu_x] \neq [t > \mu_x + v]$ . If  $s > \mu_x$  then  $t > \mu_x$  and  $s > \mu_x + v - p$ , so  $E_1$  fails. If  $s \leq \mu_x$ ,  $E_2$  forces  $t > \mu_x + v$ , so  $\mu_x + v \leq p - 2$  and  $[t > \mu_x] = 1$ , and  $E_1$  forces  $s \leq \mu_x + v - p < 0$ , impossible.

$k = 0$ : here  $\mu_y = \mu_x + 2v \leq p$ , so  $\varepsilon_y(b) = 0$  and  $E_2 \Leftrightarrow s \leq \mu_x$ ; then  $\varepsilon_y(a) = [s > \mu_x + v] = 0$  and  $E_1 \Leftrightarrow t \leq \mu_x$ , but  $t \geq p - v > p - 2v \geq \mu_x$ .

$k = 2$ : here  $\mu_y = \mu_x + 2v - 2p \geq 1$ , so  $\mu_x \geq 2p + 1 - 2v$  and  $\varepsilon_y(a) = 1$ ;  $E_1 \Leftrightarrow t > \mu_x$  and  $E_2 \Leftrightarrow (s > \mu_x) \vee (t \leq \mu_x + v - p)$ . Now  $s > \mu_x$  would give  $2p + 2 - 2v \leq s \leq p - 1 - v$ , i.e.  $v \geq p + 3$ ; so  $s \leq \mu_x$  and  $E_2$  needs  $t \leq \mu_x + v - p < \mu_x$ , contradicting  $E_1$ .

Consequently, if  $F_1$  holds then  $E_1 + E_2 + F_1 + F_2 \leq 3$ , and if  $F_1$  fails then again  $E_1 + E_2 + F_1 + F_2 \leq 3$ ; by parity the sum is 0 or 2.  $\square$

*Proof of Theorem 3.* As in (3),  $|S| \leq Z + 2|\mathcal{L}|$  for every lawful  $S \subseteq \mathcal{P}_W$ , where  $\mathcal{L}$  and  $Z$  now refer to  $\mathcal{P}_W$ . Let  $N_{sh}, N_{sp}$  be the numbers of shared and split pairs (over all  $\sigma$ - and  $\tau$ -pairs of classes with distinct partner), so  $N_{sh} + N_{sp} = (p - 1) - s$  and  $|\mathcal{L}| = N_{sh} + 2N_{sp}$ ; and  $\sum_q m(q) = \sum_{\ell \in \mathcal{L}} |\ell \cap \mathcal{P}_W| = 4N_{sh} + 6N_{sp}$ . Every point lies on one line of each slope  $\pm 1$ , so  $m(q) \leq 2$  and  $\#\{q : m(q) \geq 1\} = \sum_q m(q) - \#\{q : m(q) = 2\}$ ; hence

$$Z + 2|\mathcal{L}| = 4(p - 1) - (4N_{sh} + 6N_{sp}) + \#\{m = 2\} + 2N_{sh} + 4N_{sp} = 2(p - 1) + 2s + \#\{m = 2\}.$$

A copy  $\kappa(r, s)$  has  $m = 2$  iff both its slope-(+1) line and its slope-(-1) line are rich. If  $\kappa$  lies in a generic orbit, its copies on rich slope-(+1) lines are  $\kappa(0, 0), \kappa(1, 1)$  if the  $\sigma$ -pair is shared and these together with one of  $\kappa(0, 1), \kappa(1, 0)$  if it is split; its copies on rich slope-(-1) lines are  $\kappa(0, 1), \kappa(1, 0)$ , together with one of  $\kappa(0, 0), \kappa(1, 1)$  if the  $\tau$ -pair is split. So  $\kappa$  has exactly  $[\sigma\text{-pair split}] + [\tau\text{-pair split}]$  copies with  $m = 2$ . If  $\kappa$  lies in an exceptional orbit, no copy has  $m = 2$  (if  $\sigma\kappa = \kappa$ , the slope-(+1) lines through the copies of  $\kappa$  carry two, one and one points of  $\mathcal{P}_W$ ; similarly for the slope-(-1) lines if  $\tau\kappa = \kappa$ ). Summing over the classes,  $\#\{m = 2\} = 2 \sum_O (e_O + f_O)$  over the generic orbits  $O$  (each pair has two classes), and by Lemma 10 this is at most  $4 \cdot \#\{\text{generic orbits}\} = p - 1 - 2s$ . Therefore  $|S| \leq Z + 2|\mathcal{L}| \leq 3(p - 1)$ .  $\square$

**Corollary 11** (no-four-in-line). *In every box  $W$  the four-point rich lines are pairwise disjoint; hence the largest subset of  $\mathcal{P}_W$  with no four collinear points has exactly  $4(p - 1) - N_{sh}(W)$  points, where  $N_{sh}(W)$  is the number of shared pairs, and there are  $4^{N_{sh}(W)}$  such subsets. By the orbit lemma  $N_{sh}(W) \geq 2 \cdot \#\{\text{generic orbits}\} = \frac{1}{2}(p - 1) - s$ , so the maximum is at most  $\frac{7}{2}(p - 1) + s$ ; in the HJSW box  $N_{sh} = \frac{1}{2}(p - 1) - s$  and the set  $S_3 = H(c, p) \cap T_3$  of [2, Constr. 2.10] (with  $|S_3| = \frac{7}{2}(p - 1) + s$ ) is a largest no-four-in-line subset of  $H(c, p) \cap G(p)$ , as asserted in [2, App. A]. (No line carries five points of  $\mathcal{P}_W$ , so the no-five-in-line maximum is  $|\mathcal{P}_W| = 4(p - 1)$ .)*

*Proof.* A four-point rich line is  $D_\kappa$  for a shared  $\sigma$ -pair (points  $\kappa(0, 0), \kappa(1, 1), \sigma\kappa(0, 0), \sigma\kappa(1, 1)$ ) or  $E_\kappa$  for a shared  $\tau$ -pair (points with  $r + s = 1$ ). The slope-(-1) line through a point  $\kappa(r, r)$  is not of the second kind, and the slope-(+1) line through a point  $\kappa(r, 1 - r)$  is not of the first kind; two lines of the same kind sharing a point coincide (Lemma 4). Hence the four-point lines are pairwise disjoint, a no-four-in-line subset must miss at least one point of each of them, and deleting exactly one point from each is sufficient (all other lines carry at most three points). For the HJSW box,  $N_{sh} = \frac{1}{2}(p - 1) - s$  by Lemma 7, and  $|S_3| = |S_2| + |M \cap E \cap H(c, p)| = 3(p - 1) + \frac{1}{2}(p - 1) + s$  because the classes fixed by  $\sigma$  or  $\tau$  lie in  $E$  and the remaining classes split evenly between  $E$  and  $F$  [2, Obs. 2.9].  $\square$

For the HJSW box every generic orbit has  $e_O + f_O = 2$  (Lemma 7:  $A/D$ -orbits have  $(e, f) = (0, 2)$ ,  $B/C$ -orbits  $(2, 0)$ ), which recovers  $Z + 2|\mathcal{L}| = 3(p - 1)$ ; for the box  $[0, 2p]^2$  all pairs are shared and the maximum is  $Z + 2|\mathcal{L}| = 2(p - 1) + 2s$  (the rich lines are  $p - 1 - s$  pairwise disjoint four-point lines and there are  $4s$  free points). In general the bound  $Z + 2|\mathcal{L}|$  need not be attained (a rich three-point line may consist of points with  $m = 2$  only), and the maximum depends on the box; Table 1 shows the exact maxima for  $p = 11, c = 1$ . In all computed cases ( $p \leq 13$ , several  $c$ ) the boxes attaining  $3(p - 1)$  include the HJSW box  $x_0 = -h, y_0 \in \{0, 1\}$ ; their number depends on  $c$  (for  $p = 13$ : 28 of the 169 boxes for  $c = 1$ , 4 for  $c = 2$ ).

## 6 Beyond one hyperbola: what the theorem does not say

Theorems 1 and 3 concern a single hyperbola. Unions of two hyperbolae in the same box do exceed  $3(p - 1)$ , but — as far as we can compute — only by a few points. For  $\mathcal{P}_k = (H(1, p) \cup H(k, p)) \cap G(p)$  ( $8(p - 1)$  candidate points) exact maximum lawful subsets were computed in the project by



| $x_0 + h \setminus y_0$ | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
|-------------------------|----|----|----|----|----|----|----|----|----|----|----|
| 0                       | 30 | 30 | 30 | 28 | 26 | 24 | 22 | 24 | 26 | 28 | 30 |
| 1                       | 28 | 28 | 28 | 30 | 28 | 26 | 24 | 26 | 28 | 30 | 28 |
| 2                       | 26 | 26 | 26 | 28 | 30 | 28 | 26 | 28 | 30 | 28 | 26 |
| 3                       | 24 | 24 | 24 | 26 | 28 | 30 | 28 | 30 | 28 | 26 | 24 |
| 4                       | 22 | 22 | 22 | 24 | 26 | 28 | 30 | 28 | 26 | 24 | 22 |
| 5                       | 22 | 22 | 22 | 24 | 26 | 28 | 30 | 28 | 26 | 24 | 22 |
| 6                       | 22 | 22 | 22 | 24 | 26 | 28 | 30 | 28 | 26 | 24 | 22 |
| 7                       | 22 | 22 | 22 | 24 | 26 | 28 | 30 | 28 | 26 | 24 | 22 |
| 8                       | 24 | 24 | 24 | 26 | 28 | 30 | 28 | 30 | 28 | 26 | 24 |
| 9                       | 26 | 26 | 26 | 28 | 30 | 28 | 26 | 28 | 30 | 28 | 26 |
| 10                      | 28 | 28 | 28 | 30 | 28 | 26 | 24 | 26 | 28 | 30 | 28 |

Table 1:  $p = 11$ ,  $c = 1$ : exact maximum lawful subset of  $H(1, 11)$  in the box  $[x_0, x_0 + 22) \times [y_0, y_0 + 22)$ , indexed by  $x_0 + h \bmod p$  (rows; 0 is the HJSW position  $x_0 = -h$ ) and  $y_0 \bmod p$  (columns). Values (HiGHS MIP, witnesses through the law) range from  $2(p - 1) + 2s = 22$  to  $3(p - 1) = 30$ ; the upper bound  $Z + 2|\mathcal{L}|$  equals 30 for 76 boxes but is attained for 20.

three independent methods (SAT with two cardinality encodings, kissat and CaDiCaL; and mixed-integer programming, HiGHS; every witness verified by an independent checker):

| $p$ | $3(p - 1)$ | $k$ | max | gain | LP relaxation | method                                               |
|-----|------------|-----|-----|------|---------------|------------------------------------------------------|
| 11  | 30         | 3   | 35  | +5   | 36.00         | MIP; all $k$ : 32–35 ( $k = 2$ : 33, also SAT twice) |
| 13  | 36         | 2   | 41  | +5   | 47.54         | SAT (two encodings), MIP; all $k$ : 38–41            |
| 17  | 48         | −1  | 54  | +6   | 60.15         | MIP (11 min); $k = 2$ : 53 (MIP, 39 min)             |
| 19  | 54         | −1  | 59  | +5   | 63.64         | MIP (2 min); local search 59                         |

Three observations. (i) The double count of Section 4 does not extend: rich lines of general slope appear (a line meets each hyperbola in at most two classes, and copies of four different classes can be collinear), there are  $\Theta(p^2)$  of them, and the LP relaxation (“at most two per rich line”) is far from the truth (column “LP”), whereas for one hyperbola it is exact (Theorem 1(ii)). (ii) The gain over  $3(p - 1)$ , maximised over  $k$  (+5, +5, +6, +5 for  $p = 11, 13, 17, 19$ ; all  $k$  computed for  $p \leq 13$ ,  $k \in \{2, -1\}$  for  $p = 17$ ,  $k = -1$  for  $p = 19$ ), shows no growth. (iii) A heuristic explanation: fix the HJSW set  $S_2$  of  $H(1, p)$  and call a point  $q$  of  $H(k, p) \cap G(p)$  *blocked* if some line through  $q$  carries two points of  $S_2$ . The average number of blocking pairs per point grows like  $\log p$  (about 3.1 at  $p = 11$  and 7 at  $p = 101$  for  $k \in \{2, 3, -1\}$ ), and the number of unblocked points is 0–2 for every  $p \leq 101$  tested; freeing a point of the second hyperbola costs deletions in the first. This suggests that unions of hyperbolae in the  $2p \times 2p$  box gain  $O(1)$  (or  $o(p)$ ) over the HJSW value, i.e. that their asymptotic ratio is still  $3/2$ ; we have no proof. The same picture appears for hyperbolae with different moduli in one box: for  $H(1, p) \cup H(1, q)$  in the  $2p \times 2p$  box (each hyperbola with its own HJSW shift) the exact maxima (MIP) exceed  $3(p - 1)$  by 1, 2 or 3 in all eleven pairs computed,  $7 \leq p < q \leq 29$  — e.g. 32 vs 30 for (11, 13), 38 vs 36 for (13, 17), 50 vs 48 for (17, 19), 68 vs 66 for (23, 29) — and the number of collinear triples with points on both hyperbolae grows like  $N \log N$ , as for random sets of the same density.

## 7 Independent computer verification

The theorem has a short self-contained proof; the following check is an independent sanity test of the bookkeeping (the classification of the rich lines, the exceptional classes and the count of the maximum sets), performed by a program that trusts no part of the proof. The script `slack/hjsw.window.check.py` of the repository recomputes, from the point set alone: all lines with at least three points of  $\mathcal{P}$  (found by brute force over all pairs of points; all have slope  $\pm 1$ , three or four points, and lie inside one  $V$ -orbit); the identity  $2|\mathcal{L}| + Z = 3(p - 1)$ ;

the exact maximum by exhaustive search inside each orbit (at most  $2^{16}$  subsets), assembling a maximum witness which is passed through the independent verifier `saturation.the.law` in grid coordinates; and the number of maximum sets, compared with  $9^s$ . All checks pass for every prime  $3 \leq p \leq 101$  (all  $c \in \mathbb{F}_p^*$  for  $p \leq 31$ ;  $c \in \{1, 2, 3, -1, -2, (p+1)/2\}$  for larger  $p$ ); log: `slack/verification/hjsw_window_check_all.log`. For Theorem 3 the script `slack/hjsw_boxes_check.py` (log `slack/verification/hjsw_boxes_check.log`) checks, for every box  $(x_0, y_0 \bmod p)$  and every  $c$ : the formula  $Z + 2|\mathcal{L}| = 2(p-1) + 2s + 2\sum_O(e_O + f_O)$  against brute-force line enumeration (random boxes,  $p \leq 19$ );  $Z + 2|\mathcal{L}| \leq 3(p-1)$  for all primes  $p \leq 61$  (with equality for some boxes); the orbit lemma orbit by orbit  $((e_O, f_O) \in \{(0, 0), (0, 2), (1, 1), (2, 0)\})$  only) for all  $p \leq 31$ ; and every identity used in the proof of Lemma 10 ( $\theta = c_a + c_b - k$ ,  $\eta = c_b - c_a$ , the four split conditions, parity, and “ $F_1 \Rightarrow \neg(E_1 \wedge E_2)$ ”) on random instances up to  $p = 199$ . The exact maxima of Table 1 are MIP optima (HiGHS) with witnesses through the law. The values agree with the exact maxima obtained earlier in the project by SAT solving (kissat/CaDiCaL, two encodings, witnesses through the law): 18, 30, 36, 48, 54, 66 for  $p = 7, 11, 13, 17, 19, 23$ . An external referee report on v0.1 of this note (kept in `docs/reviews/`) re-derived the point sets and the rich lines with an independent implementation and confirmed  $3(p-1)$  and  $9^s$  for all  $c$  with  $p \leq 31$  and for samples of  $c$  with  $p = 37, 43, 59, 73, 101$ .

Repository (public): <https://github.com/iwasborninbali/saturation>, tag `paper-v0.5`; the verification script and its log were committed as `d422b78` (2026-08-17); the note itself is `paper/hjsw_window.tex`.

| $p$ | $c$ | $ \mathcal{P} $ | $ \mathcal{L} $ | $Z$ | max | #max sets |
|-----|-----|-----------------|-----------------|-----|-----|-----------|
| 7   | 1   | 24              | 8               | 2   | 18  | 9         |
| 11  | 1   | 40              | 14              | 2   | 30  | 9         |
| 13  | 1   | 48              | 16              | 4   | 36  | 81        |
| 13  | 2   | 48              | 18              | 0   | 36  | 1         |
| 17  | 1   | 64              | 22              | 4   | 48  | 81        |
| 19  | 1   | 72              | 26              | 2   | 54  | 9         |
| 23  | 1   | 88              | 32              | 2   | 66  | 9         |
| 101 | 1   | 400             | 148             | 4   | 300 | 81        |

Table 2: Sample values from the verification log ( $|\mathcal{L}| = \frac{3}{2}(p-1) - s$ ,  $Z = 2s$ ).

**AI/tool-use disclosure.** The theorem, its proof, the verification program and the text of this note were produced by an autonomous large-language-model agent (Anthropic Claude, model *Claude Fable 5*, August 2026), working in a repository shared with two earlier agents of the same kind, which had established the computational observations (exact maxima by SAT for  $p \leq 23$ ) and the setup; the human author posed the question, organised the work, obtained an external referee report, and takes responsibility for the statements made here. All code, logs and the correspondence of the agents are available in the repository.

## References

- [1] R. R. Hall, T. H. Jackson, A. Sudbery, K. Wild, Some advances in the no-three-in-line problem, *J. Combin. Theory Ser. A* 18 (1975) 336–341.
- [2] B. Kovács, Z. L. Nagy, D. R. Szabó, Randomised algebraic constructions for the no- $(k+1)$ -in-line problem, arXiv:2508.07632 (v1 11 Aug 2025, v2 21 Jul 2026); *Advances in Combinatorics* 2026:7.